Entra ID Governance: Insta-IAM/IGA Solution?

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Abstract

Organizations with complex, multi-system environments face significant challenges in managing identity lifecycle, access provisioning, and governance at scale, and typically rely on dedicated platforms or homegrown codebases to address these needs. However, the evolution of Microsoft Entra ID and its Governance capabilities raises the question of whether a standalone IGA solution is necessary for organizations already invested in the Microsoft ecosystem. This research evaluates the viability of using Entra ID Governance as a primary IAM/IGA solution through a proof-of-concept implementation modeled on a moderately complex organization with multiple systems of record. Using criteria aligned with industry-standard IGA capabilities, the study maps governance requirements to native Entra features. It tests their effectiveness across identity lifecycle management, access provisioning, reconciliation, and audit processes. Findings indicate that Entra ID Governance can fulfill core IGA functions without middleware by leveraging lifecycle workflows, access packages, and API-driven integrations. However, limitations in real-time processing, service thresholds, and attribute flexibility must be considered. The results suggest that organizations can reduce architectural complexity and accelerate adoption by consolidating identity governance within Entra ID, provided their operational requirements align with the platform's current capabilities.

1. Introduction

Organizations that rely on only a handful of systems often do not need a full Identity and Access Management or Identity Governance and Administration (IAM/IGA) platform. Consider a small retail business with 25 employees. It might operate entirely on a single system that manages inventory, sales, and finance. In this type of setup, there is often little need for system integrations or automated account provisioning.

In contrast, medium and large organizations tend to rely on multiple systems of record, support dozens of applications, and manage large populations of both employees and customers. At that scale, manual identity and access management does not hold up. According to Wessels, this growing complexity has driven the rise of the commercial IAM/IGA market, which consists of tools designed to help organizations manage it (Wessels, 2025).

One such tool is Microsoft Entra ID, which acts as the cloud-based identity and access management tool in the Microsoft 365 suite of products and Azure cloud services. Originally launched as Azure Active Directory, the Entra ID platform has steadily improved and expanded, adding capabilities that were once limited to on-premises Active Directory deployments. The planned retirement of Microsoft Identity Manager (MIM) in 2029 is the next major milestone of that shift. According to Reinders, many of MIM's advanced IAM/IGA functions are now available through Entra ID licensing tiers, including Entra ID P1/P2, Entra ID Governance, and the Entra ID Suite. As a result, complex identity problems that once required significant infrastructure and IT professionals to implement and maintain can now be achieved through cloud-based policy management (Reinders, 2025).

This research explores whether Entra ID Governance can meet the needs of a moderately complex organization without relying on a separate IAM/IGA platform. Using a focused proof of concept, it evaluates whether core identity functions can be consolidated within Entra ID, a solution already used by every company with a Microsoft cloud presence.

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2. Research Method

To evaluate Entra ID Governance's ability to address real identity and access challenges, a test environment that reflects a moderately complex organization was designed and built. The test environment includes sample events sourced from multiple identity systems of record that emulate daily operations, an Entra ID directory with Governance features enabled, and a wide range of applications and groups to which users are assigned.

2.1. Preparation

Each requirement in the test environment was examined and addressed appropriately to create a replica of a real-world information system ecosphere.

2.1.1. Systems of Record

The example organization uses multiple cloud-based Enterprise Resource Planning (ERP) systems:

- Human Resource Information System
- Student Information System
- Customer Relationship Management System
- E-Commerce System
- Finance System

The IAM/IGA test incorporated these five systems by ingesting users who require an Entra ID account from the first two and managing access to many resources, including the last three. These systems generally use modern integration techniques, such as RESTful APIs, to support automation and code-based data retrieval. To minimize costs and labor associated with setting up entire ERPs for the test environment, API payloads were sampled and anonymized from the systems above to emulate the daily events important to IAM/IGA.

2.1.2. M365 Tenant w/ Entra ID Directory

To test a Microsoft cloud directory's ability to govern user access to systems, a Microsoft 365 Tenant with Entra ID Directory is required. Although an existing Microsoft 365 tenant was available, it was determined that the safest approach for testing was to use a separate Microsoft 365 tenant. Creating a test M365 tenant generally requires contacting Microsoft, requesting a test tenant, and filling out form H-21. The required information is shown below in Table 1:

Tenant Affiliate Name (will be used as Tenant ID Name in CPS)*	**Contact First Name***	**Contact Last Name***	**Online Service Manager Email***
MaineCCTest	Scott	Fortin	mainecctest@outlook.com

Table 1: Required Information, Microsoft Tenant Request Form H-21

2.1.3. Required Microsoft Licensing

The last component that required preparation was acquiring key Microsoft licenses, which turns a common cloud directory into a powerful IAM/IGA solution. The three most important licenses involved in this research are:

Microsoft 365 A1/E1/F1 – Basic productivity licensing, which includes Exchange Online (email), online Office apps (Word Online, Excel Online, PowerPoint Online), and Entra ID Basic. This license is free to qualifying organizations.

Entra ID P1 – Add-on license with additional identity tools like multi-factor authentication, self-service password reset, single sign-on to custom applications, and application provisioning using SCIM.

Entra ID Governance – Advanced toolset that extends the ability of Entra ID to automate and govern access of member & guest users. Key features in Entra ID Governance explored by this research project include Access Packages, Lifecycle Workflows, and Access Review.

To manage the costs of the proof of concept, all three licenses were acquired through a 30-day trial period and a 30-day extension.

2.2. Criteria: Current IAM Functionality

To gauge Entra ID's ability to meet the IAM/IGA needs of a complex organization, the platform's capabilities must be evaluated against agreed-upon criteria. Gartner is in the business of examining technology solutions and uses a Market Guide to define the criteria by which multiple solutions are scored. For the IAM/IGA space, Wessler, writing for Gartner, lists the following as Mandatory Features of an IGA (Wessler, 2025):

1. Identity life cycle management and identity data integration (including with multiple sources).
2. Access request processing and workflow orchestration.
3. Access certification (also called attestation or review).
4. Provisioning via automated connectors (including some options for apps that do not use System for Cross-Domain Identity Management [SCIM]) and via integration to IT service management (ITSM)/ticketing systems to trigger manual fulfillment flows.
5. Policy and role management.
6. Auditing, reporting, and basic analytics (descriptive and diagnostic analytics), including risk scoring.
7. Entitlement management and data integration (e.g., discovery, entitlement catalog management, and entitlement data enrichment, including descriptions, owners, and sensitivity ratings).

These features and their relationship to the example organization are explored further in Appendix A. These sections are repeated in Section 3, with narrative and evidence that describe whether the research found that the Entra ID + Governance tool adequately addresses the functional area.

2.3. Proof of Concept (POC): Apply

Putting research into practice involved an iterative process to apply current functionality in the IAM/IGA system to Entra ID, creating a replica result and replacing the IAM/IGA system in part or whole.

2.3.1. Research

The POC process began with research that connected the Identity Governance and Administration (IGA) functions discussed in Section 2.2 and Appendix A with the capabilities available in Microsoft Entra ID and Entra ID Governance. This work drew on several technical sources, including Microsoft Learn documentation, TechNet articles, IAM-focused blogs, and community discussion forums. The purpose of this research was to identify the governance features most relevant to each IAM/IGA objective and to begin identifying configuration options and parameters needed to support those objectives.

2.3.2. Configuration

Once appropriate Entra ID Governance features are identified, they are validated and configured in the test environment. The configuration steps involved activating features in Entra ID, creating policies using UI wizards, and tweaking them once key parameters were identified.

2.3.3. Testing

Testing activities validated that the previously configured Entra ID features operated as expected with user data introduced both programmatically via APIs connected to an emulated system of record and manually via Entra ID configuration or PowerShell scripting. This approach allowed a wide range of identity scenarios and provisioning/deprovisioning events to be tested.

2.3.4. Analysis

In the final phase of the proof-of-concept process, the configuration and testing results were evaluated to determine whether the targeted IAM/IGA feature requirement was met. If the requirement was satisfied, the process proceeded to the next objective. IAM/IGA requirements that could not be solved or where limitations were discovered in Entra ID were noted for discussion in Section 3.

3. Findings and Discussion

The findings and results of the proof of concept are below, organized into the categories determined to be most important, as outlined in Section 2.2 and Appendix A.

3.1. Use of Entra ID / HRIS Integration features

Early in the IAM/IGA research process, the decision was made to set aside Entra ID's native HR inbound provisioning capabilities, even though they initially appeared to be a strong feature of the platform. A few practical considerations drove this choice: keeping the research scope manageable, avoiding the need to build a test HRIS environment, and, more importantly, supporting a broader architectural approach to automation in cloud-based ERP systems.

Entra ID's native HR-inbound provisioning is generally designed around tightly coupled integrations between a system of record and the Entra ID platform. According to Microsoft, Entra ID with P1 licensing supports SAP SuccessFactors & Workday HRIS out of the box. Configurable options in Entra support other platforms, but this is up to the organization or a consultant to design, configure, and test to ensure interoperability (Microsoft, 2025).

Rather than using Entra's native HR-inbound provisioning approach, the proof-of-concept environment leveraged an existing message bus and codebase to make direct Graph API calls. This process begins in systems of record, where identity-related events trigger actions: HTTPS webhooks with data payloads sent to API endpoints that execute code, standardize events, and publish messages to a bus. Middleware code receives the messages and forwards them to Entra ID for processing and storage. According to Acka, this approach helps avoid vendor lock-in, makes the solution more resilient to outages in source or destination systems, and ensures the organization can change or upgrade its upstream/downstream systems without redesigning the IAM workflow (Acka, 2026).

Although Entra's native HR & API inbound provisioning capabilities evaluated during the research phase were technically viable, the larger question was the other

IAM/IGA features available in Entra ID Governance, which act on user attributes set by either synchronization method.

3.2. Uniquely Identify Users

The Unique identifier in Entra is a GUID called 'Object ID'. In Figure 1, the main user page of Entra ID can be customized to show users' friendly names, email addresses, GUIDs, and identifiers used in an organization (Employee ID, Student ID). Object IDs are automatically assigned when a User is created- whether by automation from systems of record or by guest users from other directories.

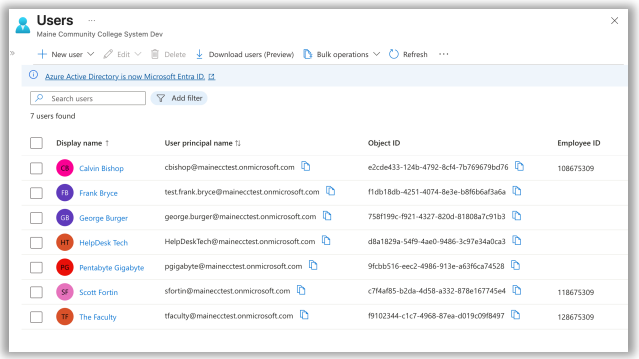


Figure 1: Unique Identifiers Displayed in Entra ID

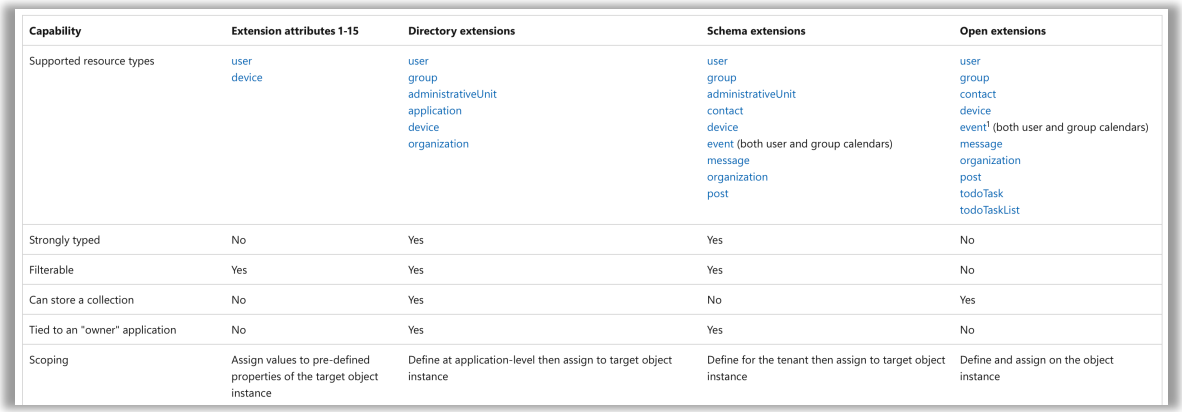
The example organization uses a numeric identifier generated by developers to de-duplicate users across multiple systems of record. This numeric identifier is stored in the Employee ID attribute in the research environment. More sensitive unique identifiers could be stored in less-visible, more-protected areas of Entra ID, as discussed further in the next section.

One important lesson learned from this research and experience in a medium-complexity organization is to use a generally accepted, widely understood, unique attribute for integrated applications. According to Cox, this identifier should be abstracted from people's data so that it persists even as people's data changes. One common example of this phenomenon is a name change, usually following marriage or other life events. The volatility of names at scale makes an identifier like email or UPN difficult to maintain. In some circumstances, email or UPN may be the only attribute a connected application can support, so procedures that compensate for this known limitation (either automated or manual) must be developed and maintained (Cox, 2025).

Other IAM/IGA products support the use of custom attributes, some of which may serve as primary identifiers, often with a moniker and broad use across an organization.

3.3. Store Attributes

This function of an IAM/IGA did not seem very complicated early in the research, given the long history of Active Directory & Entra ID performing it. However, the proof-of-concept demanded multiple iterations of experimentation to identify the correct attribute type for dependent tasks within the Entra ID suite. This experimental work can best be summarized by Figure 2, published by Microsoft, and best viewed in a web browser:



Capability	Extension attributes 1-15	Directory extensions	Schema extensions	Open extensions
Supported resource types	user device	user group administrativeUnit application device organization	user group administrativeUnit contact device event (both user and group calendars) message organization post	user group contact device event[1] (both user and group calendars) message organization post todoTask todoTaskList
Strongly typed	No	Yes	Yes	No
Filterable	Yes	Yes	Yes	No
Can store a collection	No	Yes	No	Yes
Tied to an "owner" application	No	Yes	Yes	No
Scoping	Assign values to pre-defined properties of the target object instance	Define at application-level then assign to target object instance	Define for the tenant then assign to target object instance	Define and assign on the object instance

Figure 2: Comparison of 4/5 Attribute Types in Entra ID

Early stages of research pointed towards another type of attribute not mentioned above, Custom Security Attributes. These were an attractive option, offering Graph API support, granular role-based access controls (RBAC) for sensitive data, and a neat, tidy organization into Attribute Sets. Unfortunately, Custom Security Attributes cannot be used in one key area: user application provisioning via the System for Cross-domain Identity Management (SCIM) protocol.

Based on this research, most medium- to large-sized organizations can use only three attribute types, especially those that use on-premises Active Directory. See Appendix A for a detailed explanation of these various types of attributes:

- Built-in Entra ID attributes
- Entra ID Connect-extension attributes 1-15

- Directory extension attributes

The other two attribute types available in Entra ID, schema extensions and open extensions, cannot be used in application provisioning via SCIM and other areas described in Figure 2.

Alternative IAM/IGA solutions, by nature of being abstracted from the Microsoft Entra platform, offer greater flexibility in storing and using attributes. Okta's Universal Directory is a good example of this phenomenon: User attributes are brought into the Universal Directory from one or more systems of record. Once stored, these attributes can be used for a myriad of purposes, from authentication to criteria for group membership.

3.4. Provision & Deprovision Access to Resources

3.4.1. Entra ID Enterprise Applications w/ Provisioning

One of Microsoft's core cloud directory features in Entra ID is Enterprise apps, which enable users to access and authenticate to both Microsoft online services and external applications. These external apps come in two types:

- Gallery Apps- pre-built and supported by Microsoft and the provider.
- Non-Gallery Apps- built by the organization.

These apps are further defined by the functionality they deliver, and the icons for these are shown in Figure 3 with a Gallery application:

- Federated SSO: Allows directory users to authenticate using their Entra ID credentials- usually with SAML, which transmits key user attributes 'just-in-time' to the third party.
- Provisioning: Enables deeper integration and pre-provisioning of users across third-party systems, including many attributes, group hierarchies, role assignments, and other important identity artifacts.

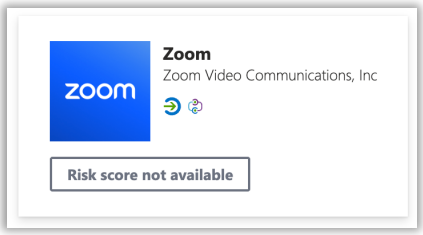


Figure 3: Zoom Gallery Application in Entra ID Enterprise Applications

To match the example organization's IAM system, app provisioning is required for both Gallery and Non-Gallery applications. A sample list is below:

- Anthology Student SIS: Non-Gallery, Provisioning Required
- Zoom: Gallery, Provisioning Required
- Rave Mobile Safety: Non-Gallery, Provisioning Required
- EZProxy Hosted: Non-Gallery, Federated SSO
- Bettermynd: Non-Gallery, Federated SSO

Only a small portion of the sample app list is handled by Gallery applications. The vast majority require non-Gallery app configurations that the organization maintains. Non-gallery federated SSO applications require the least configuration and can be implemented in a couple of hours, often with documentation provided by the third party. Non-Gallery applications that require provisioning require a more thoughtful, detailed approach to applying Microsoft's provisioning methods. According to Microsoft, several methods are supported (Microsoft, 2026):

- SCIM 2.0 direct to SaaS application
- LDAP via Entra LDAP Connector (Windows Server Application)
- SQL via Entra SQL Connector (Windows Server Application)
- Other via Entra Connect Provisioning Agent (Windows Server Application)
    - REST
    - SOAP
    - Flatfile via PowerShell (CSV export)
    - On-Prem SCIM app

An important finding of this research, on App Provisioning, relates to Section 3.2.3 on attributes. Using Entra ID as a provisioning engine for integrated applications requires that all necessary attributes be stored in Entra ID. The example organization favored SCIM 2.0 provisioning to eliminate the need for on-premises servers or installed connectors. This method requires the use of Extension Attributes or Directory Extensions, the only two custom attribute types available for SCIM provisioning, as shown in Figure 4.

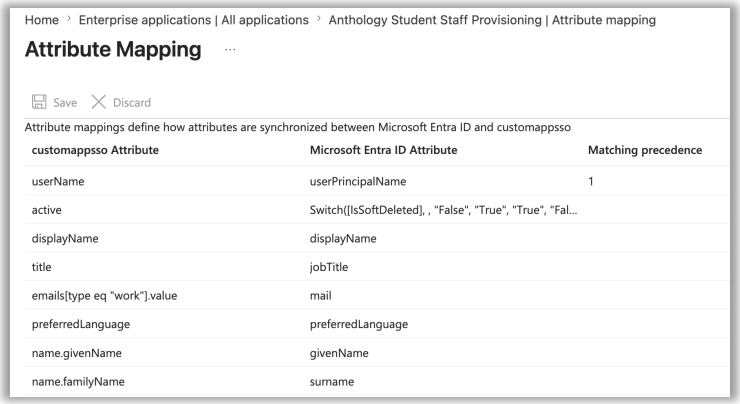


Figure 4: Enterprise App Provisioning, Attribute Mapping from Entra ID to Remote App

Bhattacharyya provides a detailed account of the use of Directory Extensions for App Provisioning, SAML Authentication, and Dynamic Groups in his blog post “How to Use Directory Extension Attributes in Entra ID for Custom Claims, SCIM Provisioning and Dynamic Groups” (Bhattacharyya, 2024).

Other IAM/IGA tools like Fischer Identity, Okta, and Sailpoint support similar provisioning methods (SCIM, API, CSV), with some of their platforms overcoming real-time limitations with true ‘immediate access’ capabilities.

3.4.2. Access Packages

Once applications are defined and built in Entra ID, the work in Entra ID Governance begins. Access Packages are bundles of resources intended to be assigned to a user or group by a Lifecycle Workflow (discussed next). According to Markdammer, resources that can be bundled into an Access Package are listed below, and an example is shown in Figure 5 (Markdammer, 2024):

- Teams
- Groups
- Applications
- SharePoint Sites
- Azure Resources (Subscriptions, Management Group)
- Entra Role
- API Permissions

Access Packages are also the vehicle through which approval chains and access review are incorporated into Entra ID, the latter of which is explored in more detail in Section 3.7.

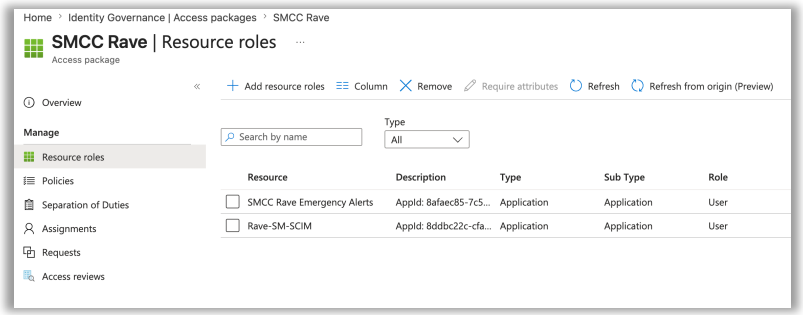


Figure 5: Access Packages with multiple applications assigned

An important service limitation in this area is the 20,000 Access Packages and 300,000 Access Package assignments per tenant (Microsoft, 2025). The latter value is important, as a large user directory will greatly limit the number of Access Packages that can be assigned to each user.

The proof of concept began with basic student and employee access packages and culminated in comprehensive resource lists organized by distinct user populations. In practice, Access Packages primarily contain Enterprise applications, as other resources can be assigned directly via a Lifecycle Workflow or Dynamic Group.

Other IAM/IGA solutions typically include this functionality under names such as “App Catalog” or “Entitlement Catalog” with advanced approval-chain customization for self-service or highly regulated environments.

3.4.3. Lifecycle Workflow Policy

Once Access Packages are used to deploy to user populations, Lifecycle Workflows combine directory knowledge about users (their attributes) with the resources they need to work or learn. Lifecycle Workflows consist of:

- Tasks: Actions (Add to group/Team, assign access package, activate/deactivate account, run Custom Task Extension) shown in Figure 6.
- Execution Conditions: Trigger (Group change, Attribute change, Time Elapsed, Sign-In Activity), Scope Details
- General Properties: Name, Schedule, Version, Category (Join/Move/Leave)

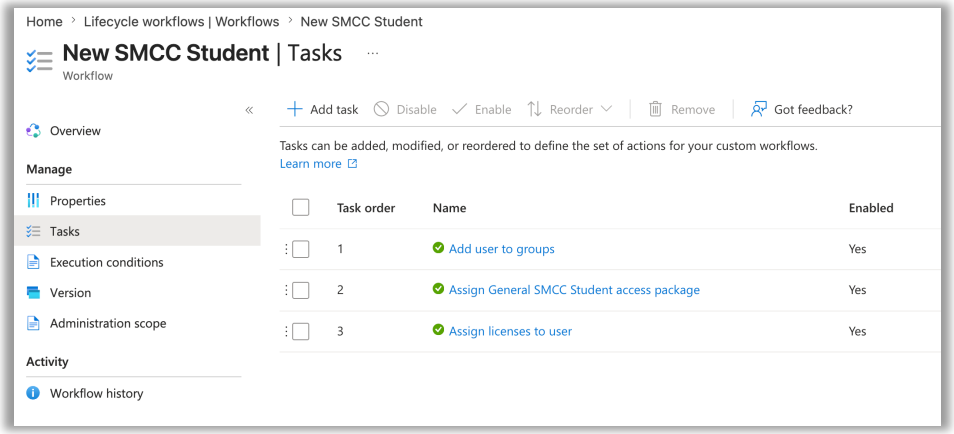


Figure 6: Lifecycle Workflow Task List

Microsoft maintains a list of important service limitations that apply to Lifecycle Workflows. The most notable limits are 100 Lifecycle Workflows per tenant, 25 Tasks per workflow, and 100 Custom Task Extensions per tenant (Microsoft, 2026).

The proof of concept started with basic Lifecycle Workflows to assign a group and an app. By the end of the research, most distinct user populations with access that can be automated and managed through policy were defined. The 100 Lifecycle Workflows service limitation was especially important in the example organization, where ‘campus’ is arguably the most important and distinguishing attribute. Each user population is multiplied by the number of campuses (7), so the example organization can have at most 12 distinct user populations per campus. The most important user populations are listed below, with 4 remaining for future growth:

- College-Employee-Staff
- Campus-Employee-Faculty
- Campus-Employee-AdjunctFaculty
- Campus-Student-General
- Campus-Student-AVD-CAD
- Campus-Student-AdobeCreativeCloud
- Campus-Student-OnlineOnly
- Campus-Student-Resident

A major finding from the proof-of-concept is the schedule associated with a Lifecycle Workflow. Even though a trigger is defined, no action will be taken until the next time a Lifecycle Workflow executes. The exception is 'On-Demand Workflows', which are triggered by manual intervention or API call. A Lifecycle Workflow schedule is set to 3 hours by default and can be reduced to 1 hour or increased to 24 hours. For an organization moving towards real-time identity actions, this may be a dealbreaker for Entra ID as an IAM/IGA.

Another challenge with Lifecycle Workflows is their interpretation of group-related events. Separate Lifecycle Workflows are required for 'Added to Group' events and 'Removed from Group' events. At scale, with multiple business units or campuses, this and other factors can quickly consume the service's 100 Lifecycle Workflows per tenant limit.

Other IAM/IGA tools offer similar functionality, such as the Lifecycle Management built into Okta, but often include real-time capabilities that ensure changes to attributes in systems of record have near-instant effects on access.

3.5. Provide End User Authentication & Identity Management Interfaces

Entra ID, formerly known as Azure Active Directory, has been around for 16 years at the time of this writing. According to Lee (2024), Microsoft 365 has over 321 million active users who regularly navigate the Entra ID/Live authentication flow. Employees or students at an organization that uses Microsoft 365 and Entra ID as its core productivity platform are even more familiar with the user experience (UX) when tools such as Multi-Factor Authentication, Self-Service Password Reset, and MyApps are used. These features are well described in other articles and are intentionally not detailed here.

End-user authentication and identity management interfaces are essential for the deeper mechanics that automate resource assignment. Organizations that adopt Entra ID Governance for IAM/IGA are highly likely to leverage these prerequisite features. Technology support divisions will need to become familiar with desktop, mobile, and

application authentication flows to provide effective support, whether in person or remotely via phone or asynchronous communication (email, chat, ticket system).

Other IAM/IGA offerings implement these features through similar user experiences and workflows. For example, Okta offers an Authenticator app, MyApps page, and Account Recovery Support to meet MFA, access, and self-service password reset requirements.

3.6. Reconcile Access After Policy Changes

Identity management problems often seem easier to solve with linear scripts than full-blown software suites. However, the real-world changes too often for a linear script to remain accurate. Changes in identity management policy are often bundled with 'cleanup work' to align pre-existing users with the new policy. Well-designed IAM/IGA systems incorporate engines that re-align users, access, and policies, typically through scheduled jobs or intervention- whether triggered or manual. Tools to review existing users with accumulated access against current policies, potentially removing or adding applications to achieve homeostasis.

In Entra ID Governance, policy changes are made to Lifecycle Workflows, which also serve as the mechanism for reconciling access. We will use an example of a new application added to an organization to demonstrate this process:

1. An institution of higher education adopts Adobe Acrobat for all faculty to improve accessibility in PDF documents.
2. Adobe Acrobat is added as an Entra ID Enterprise App- both SCIM and SSO versions for provisioning and authentication.
3. Both Enterprise Applications are added to a new Campus-Employee-Faculty-Adobe Acrobat Access Package.
4. An additional Task is added to the existing Campus-Employee-Faculty Lifecycle Workflow: Request access package: Campus-Employee-Faculty-Adobe Acrobat.

At this point, new Campus-Employee-Faculty users will receive Adobe Acrobat via the Lifecycle Workflow, but existing users will be left without it. This is where the reconciliation feature of an IAM/IGA earns its stripes. If the Campus-Employee-Faculty Lifecycle Workflow is enabled for a schedule, the next time it runs, it will determine that

most of its managed users are missing resources associated with one of its configured Tasks. The Lifecycle Workflow will execute the Tasks for all in-scope users and add the new Access Package for Adobe Acrobat.

Other IAM/IGA tools implement reconciliation features either through scheduled jobs, such as Lifecycle Workflows in Entra ID, or by detecting identity policy changes that affect many users and then iterating through those identities to ensure their access aligns with the new standard.

3.7. Access Review and Report

Access review and reporting in Entra ID Governance deviates from the often-arduous task of reviewing exported user lists and permission assignments to meet compliance or regulatory requirements. Initial configuration forces the administrator to delegate review of a particular resource (the same resources assigned in Access Packages) to owners or a group of owners. The actual review process takes place in the MyAccess area of the MyAccount self-service site, referenced in Section 3.2.4.

This paradigm indicates an access review process not relegated to IT Administrators & Security Teams. Functional owners of applications, groups, Teams, and SharePoint sites are assigned periodic reviews of their resources and are provided a modern user interface to complete them. This interface is shown in Figure 7.

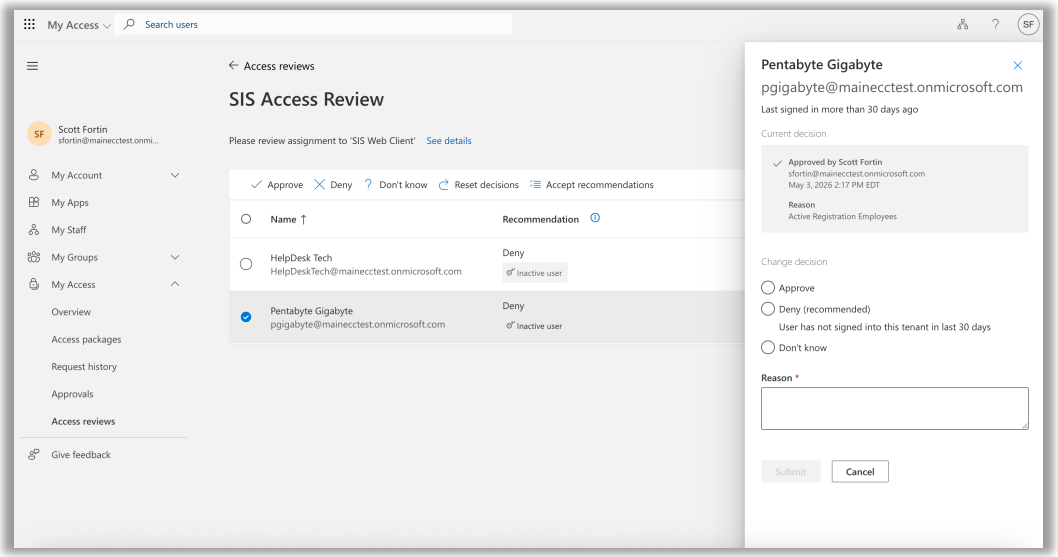


Figure 7: Access Review Results of Example Application

Other IAM/IGA tools include features very similar to Entra's Access Review and MyAccess. The main difference is that the tool is located outside the Microsoft ecosystem. Other IAM/IGA examples include Saviynt's Certification Workbench and Okta's Access Campaigns.

4. Recommendations and Implications

For organizations that already use Microsoft 365 and Entra ID, the value proposition of using existing tools to achieve IAM/IGA objectives should not be ignored. Assuming that Entra ID P1/P2 and Governance licensing are reasonably priced, the speed of adoption, reuse of existing user and group objects, and capabilities offered make for an attractive IAM/IGA solution.

Since Entra ID with Governance delivers the core capabilities typically expected of an IAM/IGA platform—as outlined in Section 2.2 and Appendix B—it can reasonably be considered a viable governance solution within the constraints identified in this study. That said, viability should not be conflated with full equivalence to traditional IGA platforms across all architectural scenarios.

In practice, whether it fits depends on the environment it is being asked to support. Organizations with a small number of well-aligned systems may find it works cleanly, while those with a broader mix of systems of record or more demanding integration needs may run into friction. Requirements around timing, workflow complexity, and the level of flexibility needed for attribute-driven provisioning- especially outside the Microsoft ecosystem- are where those differences tend to show up most clearly.

4.1. Recommendations for Practice

Organizations choosing to implement an IAM/IGA tool will arrive at this decision from diverse backgrounds and starting points. Everyone's journey will be different, but the framework outlined below includes factors that every identity engineer will encounter:

1. Determine whether real-time is required. If so, do not proceed. Entra ID uses multiple functions that run on a schedule.
2. Inventory user populations and source system-of-record, starting with the biggest and working to the smallest.
3. Inventory resources required by user populations (apps, groups, Teams, Sites).
4. Identify key attributes that indicate access to a resource from source-of-record systems.
5. Import users and key attributes into Entra ID via HRIS inbound provisioning or the Graph API.
6. Configure resources in Entra ID (Enterprise Apps, Groups, and Access Packages)
7. Build Lifecycle Workflows or Dynamic Group assignments. This step links the key attributes introduced in Entra ID to the resources required by user populations.

Depending on the organization's size and structure, Entra ID Governance service limitations with the number of Lifecycle Workflows and the number of Access Package assignments per tenant may be of concern. In this circumstance, it is highly recommended to identify further which groups and applications can be assigned through Dynamic Groups, a feature of Entra ID P1. This frees up valuable Lifecycle Workflows for more granular resource-assignment logic.

4.2. Implications for Future Research

An Entra ID feature, HRIS-inbound provisioning, which was scoped out of this research project, warrants additional research and consideration. Organizations with out-of-the-box-supported HRIS systems, small development shops, or zero-code requirements may find more value in a direct connection between an identity system of record and their IAM/IGA solution.

Other research areas generally center around getting Entra ID to participate in a real-time identity lifecycle, usually driven by demands from the business to have user access ready when they need it- sometimes on the spot. One of the areas identified as a delay to application access is Entra ID Enterprise App Provisioning. Once a user is added to an application, whether manually, by Lifecycle Workflow, or Dynamic Group, there is

a waiting period for the next provisioning cycle, which serves as the connected application. Customers should advocate for real-time, event-based functionality that triggers instant user provisioning in apps that rely on group membership.

Another feature that has the potential to improve effectiveness and reduce delays associated with scheduled Lifecycle Workflows is the On-Demand run option. Though seemingly intended to be run manually in the Entra ID UI for a high-priority, urgent app request, Microsoft also provides the ability to trigger this function via a Graph API. In theory, this function could be built into HRIS-inbound or Graph API calls when a user is provisioned, instantly evaluating the user's resource needs once their attributes are ready. Service limitations are undocumented, so engineers must assume that the standard API rate limits for Entra ID Governance also apply to this feature. Customers should advocate for expanding this feature and for clear documentation of any service limits to enable real-time execution of Lifecycle Workflows.

The final item worth mentioning in this section is the encouragement of the use of Custom Security Attributes throughout the Entra ID Suite. Microsoft has implemented a robust Custom Security Attribute feature with granular role-based access control and visibility of attributes in the Entra ID Admin Center- a major win for a technical support organization. The painful limitation of these attributes not being available for Enterprise App provisioning or determining Dynamic Group membership is a major shortfall. There exists, however, a glimmer of hope: Microsoft provides evidence that it is possible and supported to use HRIS-inbound provisioning to write to Entra ID Custom Security Attributes, enabling the reverse in the future (Microsoft, 2025).

5. Conclusion

The results of this study suggest that Entra ID Governance can serve as an effective IAM/IGA solution for organizations whose IAM/IGA needs fall within its current feature set. In the proof-of-concept environment, Entra ID provided essential functions, including attribute storage, provisioning and deprovisioning automation, reconciliation, and access reviews, all without requiring an additional third-party system.

That said, the research also revealed some clear limitations to using Entra ID Governance as an IAM/IGA solution. For organizations that depend on near real-time integrations or need to operate at a larger scale, these gaps could become significant- and in some cases, may be enough to rule it out. These limitations do not detract from what the platform does well, but they do set practical limits on where it fits.

For that reason, Entra ID Governance is not a one-size-fits-all answer to identity management challenges. Organizations already invested in Microsoft 365 and Entra will find it uniquely appealing, as it can streamline architecture, reduce overhead, and accelerate implementation by building on tools they already know and use. Still, adoption should not be automatic; organizations should take the time to inventory their IAM/IGA business requirements and exhaustively test them against prospective solutions (including Entra ID Governance) to ensure compatibility.

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Appendix A: IAM Core Features Explored

Uniquely Identify Users

In a multi-application, multi-tenant software spider web, unique identifiers are the thread that holds everything together. Though it may seem obvious, an IAM/IGA platform should include robust, tailored approaches for defining and storing unique identifiers. This becomes especially important when the IAM/IGA is responsible for connected applications and must identify users on remote systems.

True unique identifiers, such as Global Unique Identifiers (GUIDs), are difficult for humans to remember or communicate, so alternate identifiers, such as email addresses, numeric IDs, or phone numbers, are used. The ability of an IAM/IGA to index and query a person's identifiers is key in a large, complex landscape of information systems.

Store User Attributes

Most IAM/IGA tools rely on user attributes to trigger automated resource assignments. An early step in designing a proof-of-concept was to inventory all user attributes that determine access to applications and systems. These attributes were grouped into two main categories: Directory attributes and custom attributes. The first category, directory attributes, is the least sensitive and most visible to end users in familiar places like Microsoft Outlook and Teams, and includes attributes such as First Name, Last Name, and Job Title. The other category, custom attributes, contains more application-specific attributes that are generally hidden from end users. Automations use custom attributes to entitle a user to broader access. For example, an attribute specific to higher education, Student Status, contains campus affiliation, degree program, and a status (RLCC-ARTS-MATRIC). Based on this attribute:

- Groups are assigned so that a campus can target communications to audiences.
- Applications are assigned so a student can access software tools utilized by a campus, program, or course.

Provision & Deprovision Access to Applications & Groups

The next major function of the current IAM/IGA system is to gatekeep access to business applications and other resources users need to work or learn. Through attributes and policies, an IAM/IGA determines who should have access to resources, then aligns policy with configuration. The action for a given resource largely depends on the capabilities of the remote system. In modern SaaS-based systems, an IAM/IGA usually performs actions via API calls. One method gaining popularity is out-of-the-box support for user management through SCIM, the System for Cross-Domain Identity Management, which is an open standard maintained by the IETF.

Despite the availability of robust and reliable tools for user management, there are almost always older or more nuanced resources that require creative and sometimes painful methods to get users in or out and assign the appropriate level of access: Full or incremental CSV exports, SQL inserts/deletes, and worst case- manual action.

Provide End User Authentication & Identity Management Interfaces

Much of what makes an IAM/IGA system valuable is delivered to IT professionals and hidden from end users, such as employees, students, or customers. The parts visible to end users typically include authentication challenge pages when accessing an application and a self-service portal where users can manage their identity. Some identity management functions are restricted- like allowing end users to change their email address. Other functions should involve end users- like empowering a user to change their password if they suspect it has been compromised, managing multi-factor authentication devices, security questions, and even evolving attributes like preferred name.

Microsoft 365 & Entra ID include these well-documented, familiar features. The example organization actively uses these features; as such, the proof-of-concept and research will largely exclude them from testing.

Reconcile Access After Policy Changes

Policies that dictate which users have access to which resources change. After a policy change, an organization without an IAM/IGA must manually update app and group assignments to align with the new policy. As you can imagine, this is monotonous,

error-prone, and difficult at scale. A core function of a robust IAM/IGA tool is to reconcile user access across an ever-changing policy landscape. User accounts are reviewed periodically, and actions are taken in managed applications to ensure users have just the right amount of access.

Access Review and Report

Most organizations are subject to information security regulations, and almost every such regulation or framework includes requirements for identity and access management. Demonstrating compliance with these required practices typically involves reviewing user access to regulated systems. An organization's IAM/IGA platform is uniquely positioned to make this process easier if it includes useful reporting, compliance, and attestation features.

Appendix B: Gartner Mandatory Feature to Entra ID Governance Crosswalk

Gartner Market Guide IGA Feature	**Entra ID Governance Feature**
1. Identity life cycle management and identity data integration (including with multiple sources).	HRIS-inbound provisioning, Graph API extensibility, Lifecycle Workflows
2. Access request processing and workflow orchestration.	Access Packages, Access Requests
3. Access certification (also called attestation or review).	Access Review
4. Provisioning via automated connectors (including some options for apps that don't use System for Cross-Domain Identity Management [SCIM]) and via integration to IT service management (ITSM)/ticketing systems to trigger manual fulfillment flows.	Enterprise Apps Provisioning, Entra Connectors (LDAP, SQL, Provisoning Agent)
5. Policy and role management.	Privileged Identity Management, Identity Governance Administrator role, Lifecycle Workflows
6. Auditing, reporting and basic analytics (descriptive and diagnostic analytics), including risk scoring.	Audit Log, Identity Governance Status, Risky Users
7. Entitlement management and data integration (e.g., discovery, entitlement catalog management, and entitlement data enrichment, including descriptions, owners and sensitivity ratings).	Entitlement management: Catalogs, My Access portal

Appendix C: Entra ID Attribute Types

Directory Attributes

Certain attributes are predictable in a directory- first name, last name, email, job title, department, phone numbers, and more. Microsoft has been storing these attributes in a central location since the eighties. Today, these same attributes are stored in Entra ID, Active Directory, or both with the Entra Connect sync tool.

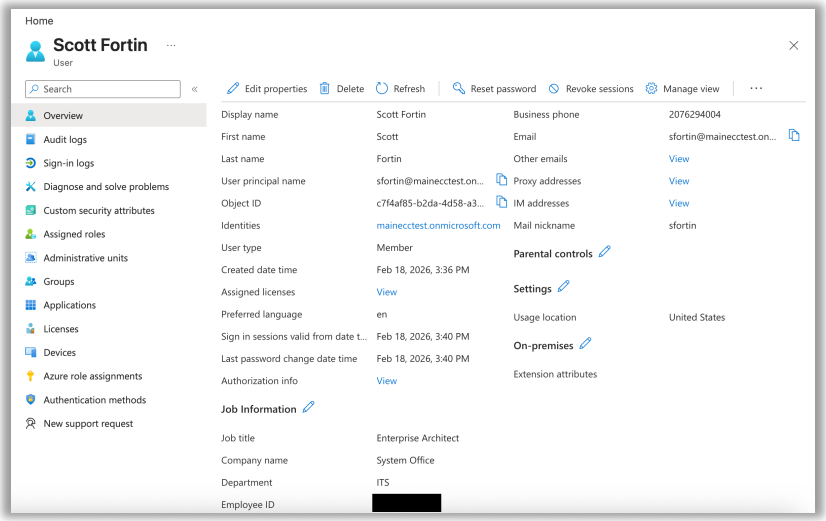


Figure 8: Directory Attributes viewed in Entra ID

No matter how many attributes an IAM/IGA predefines for organizations, custom security attributes will inevitably be required. Below are attribute types that can be adapted for a variety of uses by an organization.

Extension Attributes 1-15

Early in the evolution of Azure Active Directory (now Entra ID), a handful of Extension Attributes were made available for organizations to sync between their tried-and-true on-premises Active Directory and their new cloud presence. According to Ayilliath, organizations can enable sixteen Extension Attributes for use in Entra ID and on-premises Active Directory. These attributes are usable in Dynamic Groups, SCIM

provisioning, SAML claims, Lifecycle Workflow attribute triggers, and more. (Ayilliath, 2024).

Directory Extensions

For organizations with custom attribute requirements that exceed the 16 Extension Attributes available, Microsoft introduced Directory Extensions specific to Azure Active Directory (now Entra ID). Directory Extensions are associated and stored within a separate Enterprise App and an object type (user, device, etc). According to Microsoft, a tenant may have 100 directory extensions per resource group (Microsoft, 2025).

Using Directory Extensions in Entra ID requires the Enterprise App ID (GUID format) to locate the object where user attributes are stored. Once located, the attributes can be used in Dynamic Groups, SCIM provisioning, SAML claims, Lifecycle Workflow attribute triggers, and more.

Custom Security Attributes

Entra ID allows organizations to define custom Attribute Sets and custom Attributes within those sets. An important limitation discovered as part of this research is the maximum number of custom Attribute Sets- 500, and the maximum number of active custom Attributes- 500 (Microsoft, 2025).

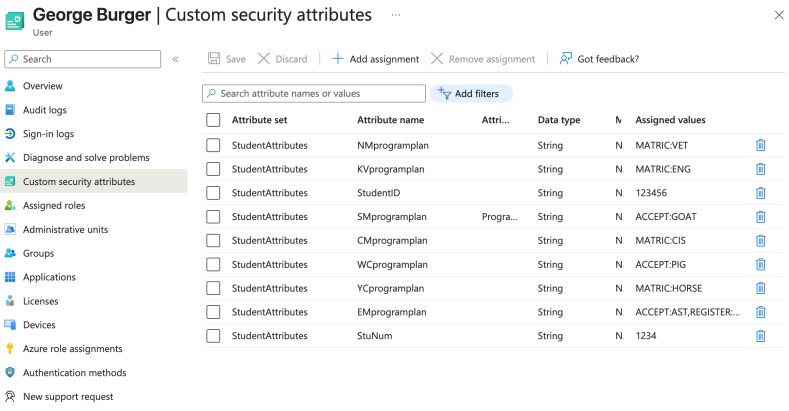


Figure 9: User Custom Security Attributes Viewed in Entra ID

While not immediately evident, Attribute Sets are an important feature for organizing custom security attributes and enforcing granular permissions to read or

modify them. The example organization utilizes about 20 custom attributes split into two major sets: EmployeeAttributes and StudentAttributes that originate from systems of record: SIS and HRIS, per campus. Seven campuses in the system yield about 140 custom attributes- 28% of the maximum.

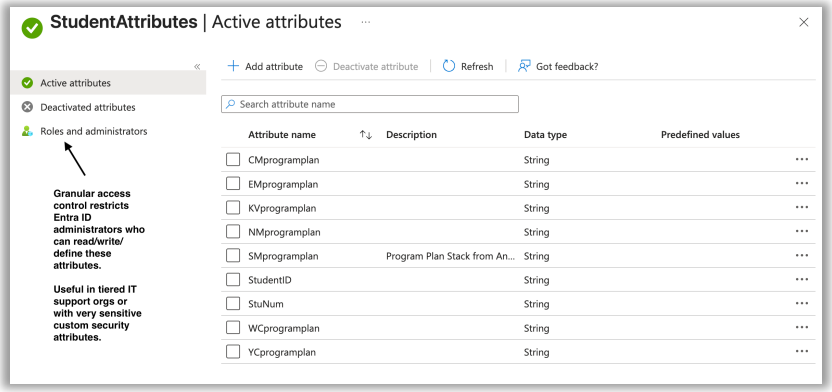


Figure 10: User Custom Security Attributes Viewed in Entra ID

Custom Security Attributes were eventually eliminated from contention as a viable solution for the proof of concept because they cannot be used for Enterprise App provisioning via SCIM or other methods. If this limitation is resolved in the future, the other benefits of Custom Security Attributes will make them an attractive option for storing key attributes for identity automation, management, and governance.

Other Attributes

Two other attribute types available in Entra ID, Schema Extensions, and Open Extension, have Microsoft documented limitations for use in key areas like app provisioning, dynamic groups, and Lifecycle Workflows, and were excluded from research early on (Microsoft, 2025).